Home Assistant with Zigbee ZHA

Data flow diagram sheets | Revision 1.0 | Sheet 1 of 3

Every flow is drawn once. Repeated IDs denote the same element. Arrows include associated responses.
Boundary labels identify crossed trust boundaries defined in the companion package.

External entity | Process | Data store | Dashed frame = flow grouping

DF1 Boundary TB4

EE1 Household administrator; EE2 Ordinary household member

Credentials and dashboard actions; local user input →

P1 Frontend on browser or phone

DF2 Boundary TB1,TB4

P1 Frontend on browser or phone

Frontend load login REST and WebSocket calls plus tokens; HTTP and WS →

P2 Core APIs authentication state machine and event bus

DF3 Boundary TB1,TB4

EE6 Local script holding long lived token

Script service calls with long lived token; local HTTP →

P2 Core APIs authentication state machine and event bus

DF4 Boundary none

P2 Core APIs authentication state machine and event bus

Events service calls and state; in process →

P3 Automation engine inside Core

DF5 Boundary none

P3 Automation engine inside Core

Automation commands; in process →

P2 Core APIs authentication state machine and event bus

DF6 Boundary none

P2 Core APIs authentication state machine and event bus

Device commands and returned state; in process →

P4 ZHA inside Core

DF7 Boundary TB2

P4 ZHA inside Core

Coordinator commands and Zigbee payloads; USB serial →

P5 USB coordinator firmware

P = process EE = external entity DS = data store TB = trust boundary DF = data flow

Logical view; process boxes do not imply isolation. Full scope and controls are specified in the package.

Home Assistant with Zigbee ZHA

Data flow diagram sheets | Revision 1.0 | Sheet 2 of 3

Every flow is drawn once. Repeated IDs denote the same element. Arrows include associated responses.
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External entity | Process | Data store | Dashed frame = flow grouping

DF8 Boundary TB2

P5 USB coordinator firmware	Commands reports and joining frames; Zigbee radio →	EE3 Zigbee routers smart plug and bulb; EE4 Zigbee end devices door and motion sensors

DF9 Boundary none

EE3 Zigbee routers smart plug and bulb	Mesh relay traffic; Zigbee radio →	EE3 Zigbee routers smart plug and bulb; EE4 Zigbee end devices door and motion sensors

DF10 Boundary TB1,TB4

EE1 Household administrator	60 second pairing request via P1 and P2; HTTP then in process →	P4 ZHA inside Core

DF11 Boundary TB3

EE5 Manufacturer OTA provider	Firmware and metadata fetched by ZHA; outbound HTTPS request and response →	P4 ZHA inside Core

DF12 Boundary none

P2 Core APIs authentication state machine and event bus	Configuration account and token record reads and writes; local storage →	DS1 Configuration and integration secrets; DS2 Accounts hashes and token validation records

DF13 Boundary none

P4 ZHA inside Core	Network keys and device state reads and writes; local storage →	DS3 Zigbee keys device list and network state

DF14 Boundary none

P2 Core APIs authentication state machine and event bus	State and event history writes; local database →	DS4 Recorder state and event database

P = process EE = external entity DS = data store TB = trust boundary DF = data flow

Logical view; process boxes do not imply isolation. Full scope and controls are specified in the package.

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Data flow diagram sheets | Revision 1.0 | Sheet 3 of 3

Every flow is drawn once. Repeated IDs denote the same element. Arrows include associated responses.
Boundary labels identify crossed trust boundaries defined in the companion package.

External entity | Process | Data store | Dashed frame = flow grouping

DF15 Boundary none

P7 Backup manager inside Core	Read persistent state to create scheduled backup; local storage →	DS1,DS2,DS3,DS4 | Data stores

DF16 Boundary none

P7 Backup manager inside Core	Write encrypted archive and manage encryption key; local storage →	DS5 Encrypted local backup archive; DS6 Backup encryption key material

DF17 Boundary none

P6 Custom integration inside Core	In process calls and access with Core privileges →	P2,DS1,DS2,DS3,DS4,DS5,DS6,DS7 | See component inventory

DF18 Boundary none

P2,P4,P7 | See component inventory	Authentication integration and backup events; local writes →	DS7 Local application logs

P = process EE = external entity DS = data store TB = trust boundary DF = data flow

Logical view; process boxes do not imply isolation. Full scope and controls are specified in the package.